SIMATS ENGINEERING

Secure Exit

SIMATS Online Programming Lab

JAVA PROGRAMMING

Jeyakannan L
192525376RD

CO3-AT2-LEVEL EXAM 2

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QUESTIONS

LEVEL 2 — MEDIUM

- **Q1** Array
- **Q2** Numbers
- **Q3** Control
- **Q4** Strings

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Level 2 — Medium **(M) Matrix Mult.** — Array

```
Write a program for matrix multiplication?
Sample Input:
Mat1 =
1  2
5  3
Mat2 =
2   3
4   1
Sample Output:
Mat Sum =
10    5
22    18
```

Your Code

```
import java.util.Scanner;
class a{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int m = sc.nextInt();
        int [][] a = new int[n][m];
        int [][] b = new int[n][m];
        for(int i=0;i<n;i++){
            for(int j=0;j<m;j++){
                a[i][j]=sc.nextInt();
            }
        }
        for(int i=0;i<n;i++){
            for(int j=0;j<m;j++){
                b[i][j]=sc.nextInt();

            }
        }
        int [][] result = new int[n][m];
        for(int i=0;i<n;i++){
            for(int j=0;j<m;j++){
                result[i][j]=0;
                for(int k=0;k<n;k++){
                result[i][j]+=a[i][k]*b[k][j];
            }
            }
        }
        System.out.println("Mat Sum =");
        for(int i=0;i<n;i++){
```

```
            for(int j=0;j<n;j++){
                System.out.print(result[i][j]+" ");
            }
            System.out.println();
        }
    }
}
```

Program Input (stdin)

```
2
2
1
2
```

Output

```
Mat Sum =
10 5
22 18
```

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- Q4 Strings

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Level 2 — Medium **(M) Prime and Composite** — Numbers

```
Write a program to count all the prime and composite numbers entered by
the user.
Sample Input:
Enter the numbers
4
54
29
71
7
59
98
23
Sample Output:
Composite number:3
Prime number:5
```

Testcases:

```
1.      33, 41, 52, 61,73,90
2.      TEN, FIFTY, SIXTY-ONE, SEVENTY-SEVEN, NINE
3.      45, 87, 09, 5.0 ,2.3, 0.4
4.      -54, -76, -97, -23, -33, -98
5.      45, 73, 00, 50, 67, 44
```

Your Code

```java
import java.util.Scanner;
class a{
    public static void main(String[] args){
    Scanner sc = new Scanner(System.in);
    int n = sc.nextInt();
    int prime=0;
    int composite=0;
    for(int i=0;i<n;i++){
         int a = sc.nextInt();
         int count =0;
         for(int j=1;j<=a;j++){
             if(a%j==0){
                 count++;
               }
           }
             if(count==2){
                 prime++;
              }
                else
                composite++;
             }

```

```
    System.out.println("Composite number : "+composite);
    System.out.println("Prime number : "+prime);
    }
}
```

Program Input (stdin)

```
8
4
54
29
```

Output

```
Composite number : 3
Prime number : 5
```

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- **Q1** Array
- **Q2** Numbers
- **Q3** Control
- **Q4** Strings

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Level 2 — Medium **(M) Tax** — Control

```
Write a program to calculate tax given the following conditions:
a.      If income is less than or equal to 1,50,000 then no tax
b.      If taxable income is 1,50,001 – 3,00,000 the charge 10% tax
c.      If taxable income is 3,00,001 – 5,00,000 the charge 20% tax
d.      If taxable income is above 5,00,001 then charge 30% tax

Sample Input:
Enter the income:200000
Sample Output:
Tax= 20000
```

Testcases:

```
Test cases:
1.      400700
2.      2789239
3.      150000
4.      00000
5.      -125486
```

Your Code

```java
import java.util.Scanner;
class a{
    public static void main(String[] args){
   Scanner sc = new Scanner(System.in);
   double income = sc.nextInt();
   double tax=0;
   if(income<=150000){
      tax=0;
     }
  else if(income>=150001&&income<=300000){
    tax=income*0.10;
}
  else if(income>=300001&&income<=500000){
     tax=income*0.20;
        }
  else if(income>=500001){
     tax=income*0.30;
        }
   System.out.println("Tax = "+tax);



    }
}
```

Program Input (stdin)

```
200000
```

Output

```
Tax = 20000.0
```

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- **Q1** Array
- **Q2** Numbers
- **Q3** Control
- **Q4** Strings

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Level 2 — Medium **(M) Palindrome** — Strings

```
Write a program using choice to check
Case 1: Given string is palindrome or not
Case 2: Given number is palindrome or not
Sample Input:
Case = 1
String = MADAM
Sample Output:
Palindrome
```

Testcases:

```
1. MONEY
2. 5678765
3. MALAY12321ALAM
4. MALAYALAM
5. 1234.4321
```

Your Code

```java
import java.util.Scanner;
class a{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        if(n==1){
            String s = sc.next();
            String rev="";
            for(int i=s.length()-1;i>=0;i--){
                rev=rev+s.charAt(i);
            }
            if(rev.equals(s)){
                System.out.println("Palindrome");
            }
            else{
                System.out.println("Not palindrome");
            }
        }
        else if(n==2){
            int a = sc.nextInt();
            int rem;
            int revn=0;
            int temp =a;
            int base=1;
            while(temp>0){
                rem=temp%10;
                revn=revn+rem*base;
                base=base*10;
```

```java
                temp=temp/10;
            }

            if(revn==a){

                System.out.println("palindrome");
            }
            else{
                System.out.println("Not palindrome");
            }
        }
        else{
            System.out.println("Invalid choose");
        }
    }
}
```

Program Input (stdin)

```
1
madam
```

Output

```
Palindrome
```

Run Save Submit

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